Cooling System

Control of engine temperature is accomplished by means of a thermostat directing the flow of cooling liquid within a sealed cooling system. A sealed system utilizes a pressure valve which maintains a slight pressure within the system when engine is running at normal operating temperature, thus raising the efficiency of the system. A condensing tank or surge tank, permits expansion of cooling liquid without loss from the system.

Cooling liquid is circulated by means of a centrifugal pump driven from the blower. Pump draws cooling liquid through radiator and oil cooler, discharging it into lower part of cylinder block. Openings in water jackets around cylinder bores connect with corresponding openings in cylinder head, circulating liquid around exhaust valves and fuel injectors. A water manifold bolted to cylinder head, discharges cooling liquid into radiator where it is then cooled by action of fan.

During the engine warm-up period, action of thermostat closes off flow of cooling liquid into radiator, directing the flow of cooling liquid through a by-pass tube. This arrangement permits circulation of liquid within the engine warming the engine and shortening the warm-up period. As engine reaches normal operating temperature, thermostat valve opens gradually, allowing cooling liquid to flow through radiator, thus maintaining an efficient operating temperature.

Thermostatically controlled radiator shutters automatically assist engine thermostat in controlling engine operating temperature.

Refer to applicable vehicle maintenance manual for specific, thermostat and radiator shutter information; also for information regarding other cooling system units as used on specific vehicles.

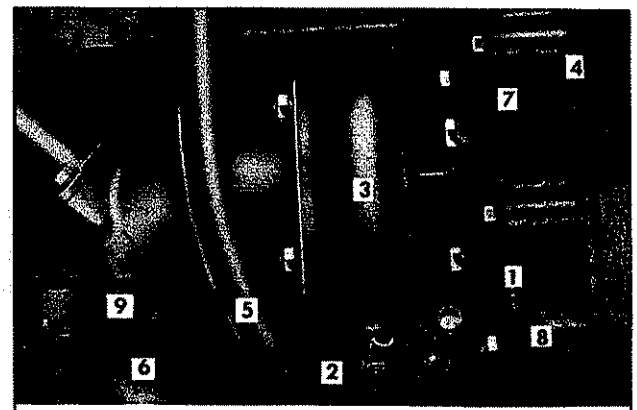
WATER PUMP

DESCRIPTION

A centrifugal type water pump (fig. 2) is used for circulating the cooling liquid through the engine. A bronze impeller with straight blades is pressed and pinned onto one end of a case-hardened steel shaft, and a pump drive coupling and oil slinger is pressed onto the opposite end. Oil slinger shrouds inner end of pump body flange to prevent oil from seeping along shaft and through shaft bearing. Shaft is supported at drive end on a sealed, double-row combination radial and thrust ball bearing. Shaft is prevented from moving endwise by staking pump housing at the inner end of the bearing.

A spring loaded carbon and Neoprene seal, retained in impeller by a steel stamping, prevents water from seeping along shaft from impeller end. The slinger is fitted on shaft between pump housing and ball bearing to prevent moisture from creeping along shaft to the bearing. As shaft serves as inner race of ball bearing pump shaft, bearing and slinger constitute one assembly and are serviced as such.

Water pump ball bearing is of the "Sealed" type and filled with lubricant when assembled, requiring no further lubrication.



1 Flange - Outlet Packing
2 Drain Cock
3 Body - Water Pump
4 Blower End Plate
5 Inlet Seal
6 Housing - Oil Cooler
7 Bolt - Water Pump to Blower
8 Bolt - Packing Flange to Cylinder Block
9 Hose Clamp TPM-3021

Figure 1—Water Pump Installed—Typical

REMOVAL

Refer to figure 1 and remove water pump in the following manner:

1. Drain cooling system.